SigNova: Bridging Communication with Sign Language Translation

By: Alshimaa Ahmed Mohammed, Farha Ahmed Ali, Fagr Ashraf Ismael, Bansee Abdelnasser Mahmoud, Omnia Mahmoud Ahmed, Nada Tarek Saber

Supervised By: Dr. Yasmine Afify , AL. Rezq Mohammed

Information Systems Department

Faculty of Computer & Information Sciences - Ain Shams University

1 Introduction

- Communication between Deaf and hearing individuals remains challenging due to distinct language modalities. Traditional reliance on scarce human interpreters highlights an urgent need for scalable digital solutions.
- Recent advancements in AI, Computer Vision, and NLP offer new pathways to bridge this gap. Can an integrated, bidirectional mobile application effectively eliminate real-time communication barriers?
- SigNova addresses the critical limitations of existing single-direction tools by focusing on American Sign Language (ASL), offering an all-in-one platform featuring automated Sign-to-Text translation, Text-to-Sign visualization, and real-time interactive messaging.

Our Objectives

01 Resolve Challenges for the Deaf Community

02 Build Accessible Technology

03 Innovate Communication Methods

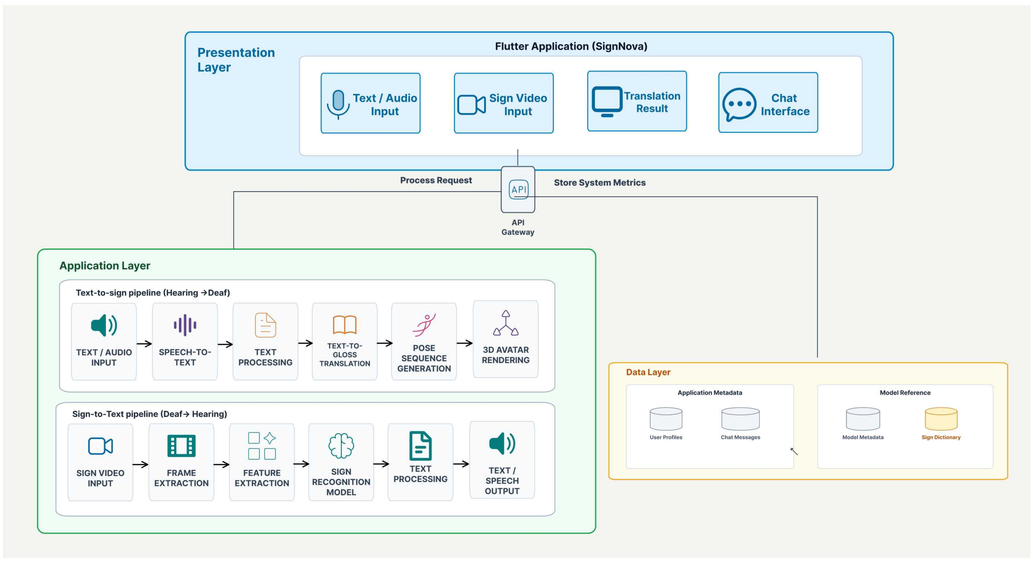

Figure 1 : System Architecture

2 Methodology

SigNova uses two AI pipelines for real-time bidirectional translation between Sign Language and English :

- The Sign-to-Text pipeline extracts hand, face, and body landmarks using MediaPipe, then uses an I3D model trained on the WLASL dataset to recognize sign sequences and convert them into English text.
- The Text-to-Sign pipeline uses Groq's LLaMA 3.1 model to generate ASL gloss from input text, which is mapped to a sign landmark database to produce animations, with fingerspelling as a fallback for unknown words.
- Together, these pipelines enable accurate and seamless communication between deaf and hearing users.

Text-to-Sign Pipeline

English text is processed using Groq's LLaMA 3.1 model to generate ASL gloss. The gloss is mapped to a sign landmark database to produce structured sign animations, with fingerspelling used for unknown words.

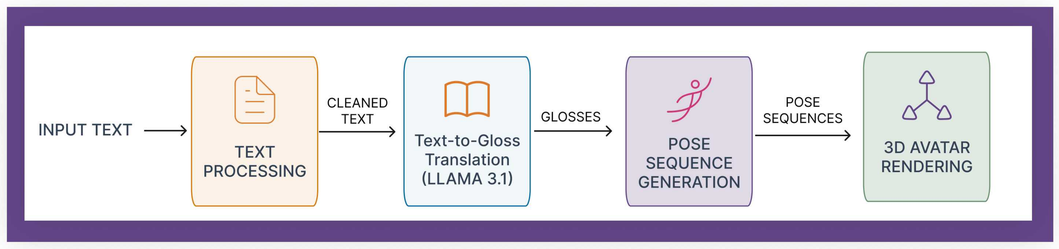

Figure 2 : Text-to-sign Pipeline and Methods Used

Sign-to-Text Pipeline

MediaPipe extracts hand, face, and body landmarks from input video. These features are processed by an I3D deep learning model trained on the WLASL dataset to recognize sign glosses, which are then converted into English sentences.

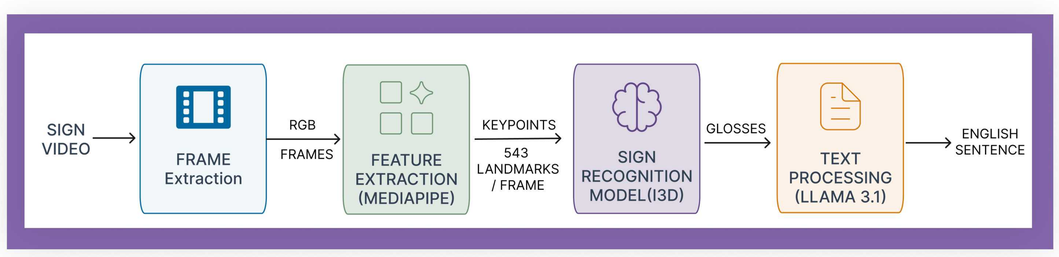

Figure 3 : Sign-to-text Pipeline and Methods Used

3 Key Algorithms

SigNova integrates advanced computer vision and artificial intelligence techniques to enable accurate bidirectional translation between sign language and text. The core algorithms used in each pipeline are summarized below.

Sign-to-text

01 **MediaPipe Holistic:** Extracts body, hand, and pose landmarks from video frames.

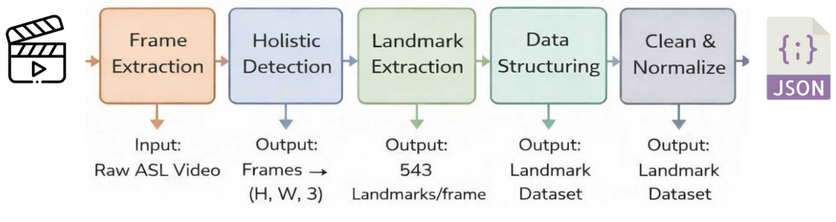

Figure 4 : MediaPipe Holistic Process

02 **I3D (Inflated 3D ConvNet):** Learns spatiotemporal features for sign recognition.

Why I3D?

- Video-based action recognition, capturing both spatial and temporal features.

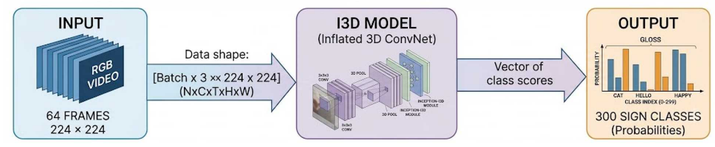

Figure 5 : I3D model inference parameters

03 **TOP-k Predictions :** Top-K Prediction helps the system choose the most confident sign(s) and provides multiple alternatives when needed.

Figure 6 : Top-1 , TOP-5 , TOP-10 Predictions

Text-to-sign

01 **MediaPipe Holistic:** Extracts body, hand, and pose landmarks from video frames to build the sign database.

02 **LLAMA 3.1 (Groq) :** LLM-Based Gloss Generation convert English text into ASL glosses

03 **Gloss Mapping:** Maps processed text to corresponding sign language glosses.

04 **Video Retrieval:** Retrieves and sequences pre-recorded sign videos to generate the final output.

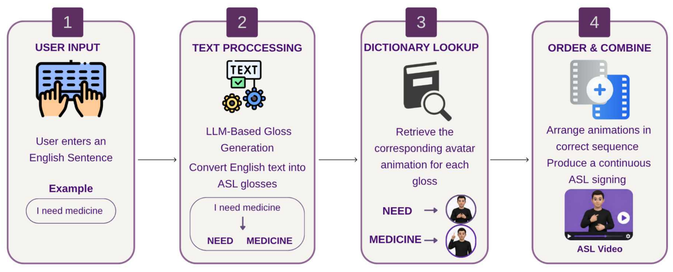

Figure 7 : LLM-Driven 3D Avatar Translation

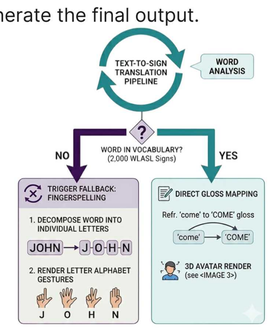

Figure 8 : Fullback Strategy

4 Results

Sign-to-text Pipeline Performance

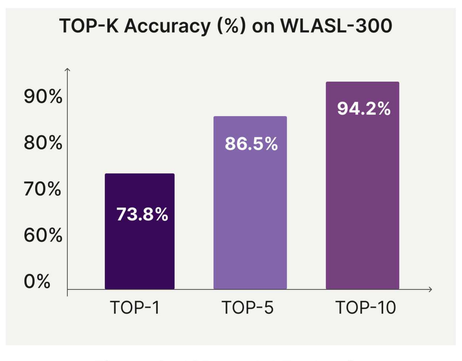

Figure 9 : I3D model Evaluation

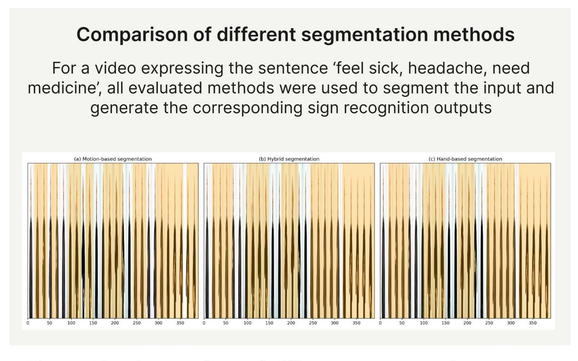

Figure 10 : Comparison of different segmentation methods

Text-to-sign Pipeline Performance

T5-Based model	LLAMA 3.1
Strong on simple sentences.	High contextual understanding.
Weak generalization on complex inputs.	Stable performance across sentence types.
Sensitive to dataset noise.	More natural and coherent gloss generation.
Inconsistent structural output.	Better alignment with intended meaning.

Table 1 : Comparison of different segmentation methods

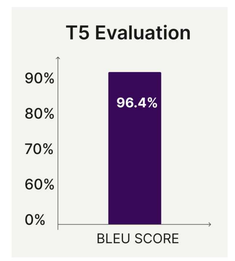

Figure 11 : T5 Evaluation

Overall System Performance

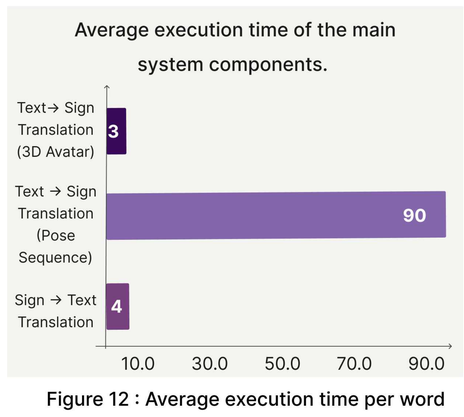

Figure 12 : Average execution time per word

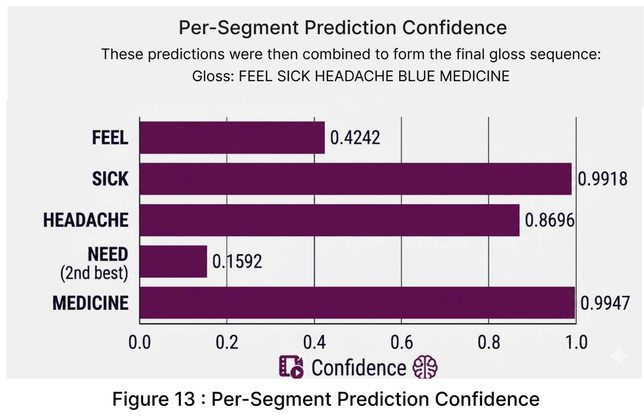

Figure 13 : Per-Segment Prediction Confidence

5 Conclusion

SigNova addresses communication barriers between deaf and hearing individuals through an AI-powered platform that combines computer vision, natural language processing, and mobile technologies. It provides bidirectional translation between English text and American Sign Language (ASL), along with an integrated chat system for seamless communication. By unifying these features in a single application, SigNova enhances accessibility and improves the user experience. The system leverages pretrained deep learning models and a scalable architecture to ensure accurate and efficient translation. It also provides a strong foundation for future intelligent assistive technologies, promoting digital inclusion and more effective communication between deaf and hearing communities.

6 References

[1] W. F. Maia, A. M. Lopes, and S. David, "Automatic Sign Language to Text Translation Using MediaPipe and Transformer Architectures," Neurocomputing, vol. 642, Art. 130421, 2025.

[2] S. H. Arib, R. Akter, S. Rahman, and S. Rahman, "SignFormer-GCN: Continuous Sign Language Translation Using Spatio-Temporal Graph Convolutional Networks," PLOS ONE, vol. 20, no. 2, 2025.

[3] A. Wang, J. Li, W. Luan, and L. Pan, "OVSLT: Advancing Sign Language Translation with Open Vocabulary," Electronics, vol. 14, no. 5, 2025. S. Fang, L. Wang, C. Zheng, Y. Tian, and C. Chen, "SignLLM: Sign Languages Production Large Language Models," arXiv preprint arXiv:2405.10718, 2024.

[4]J. Gong, L. G. Foo, Y. He, H. Rahmani, and J. Liu, "LLMs are Good Sign Language Translators," arXiv preprint arXiv:2404.00925, 2024.